Nicolo DiFerdinando

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About Me

Industrial engineering graduate with experience in software engineering, data engineering, and cybersecurity automation. Focused on building internal tools that streamline workflows, designing data pipelines and database systems, automating reporting and repetitive tasks, and building agents with Langchain and Python.

Education

Bachelor of Science in Industrial Engineering *Sept 2021 – June 2026*
California Polytechnic State University, San Luis Obispo
Coursework: Data Science, Computer Science, Data Analytics, Enterprise Analytics, Database Management & System Design, Operations Research, Machine Learning, Statistics & Design of Experiments, Simulation, Human Factors & Ergonomics, Usability / HCI, Project Management, Internal Systems Management, RAG-assisted LLM systems.

Experience

- **AI-Driven Solutions Analyst (Full-time)** – *PeopleConnect* *Aug 2026 – Present*
  - **AI Automation & Tooling:** Build and deploy LLM-powered tools and Python scripts using OpenAI, Claude, Zapier, and Airflow to automate high-impact workflows across Growth, BI, and Operations.
  - **Growth & Analytics Enablement:** Partner with Marketing and Finance on creative generation and campaign optimization, applying ML and prompt-engineered systems to improve forecasting across Snowflake, dbt, and Sigma.
  - **Cross-Functional Impact:** Translate business problems into AI-driven solutions. Run structured experiments and measure outcomes on time savings, revenue lift, and cost efficiency to scale what works.
- **Business Intelligence Intern (Part-time)** – *PeopleConnect* *Oct 2025 – June 2026*
  - **Dashboard Development:** Build interactive dashboards using Sigma Computing, Tableau, and Snowflake serving 50+ stakeholders across Growth Operations, Executives and Finance through cross-functional collaboration.
  - **Data Pipeline Engineering:** Develop and monitor data pipelines using dbt core/fusion semantic layers with CI/CD workflows; manage ETL pipelines for incremental logic with code review processes across the data engineering team.
  - **Semantic Layer Service:** Built a service embedding company metrics, dimensions, and entities over the dbt semantic layer, exposing governed definitions to the interface of choice through MCP connectors (Claude, ChatGPT), a Slack bot, and more.
  - **Reporting Migration:** Assisted in migrating enterprise reporting from Tableau to Sigma Computing, streamlining dashboard delivery and reducing licensing overhead.
- **Cybersecurity & IT Intern (Full-time)** – *PeopleConnect* *June 2025 – Sept 2025*
  - **SOC 2 Compliance:** Developed bash scripts automating audit reporting workflows, completing SOC 2 audit in 4 weeks vs. projected 10; communicated findings to engineering and compliance stakeholders.
  - **Enterprise Security:** Deployed vulnerability scanning and testing solution for 200+ endpoints; created reusable automation scripts allowing further visibility into on prem devices, reducing risk surface.
- **Assistant Student SOC Analyst (Part-time)** – *Cal Poly ITS* *April 2024 – June 2025*
  - **Security Automation & Incident Response:** Built a Python Slack app automating SOC workflows (email confirmation, user suppression, blocking) by integrating Splunk SOAR playbooks with Slack Bolt REST APIs for real-time incident response, and analyzed and responded to 50+ weekly security incidents (suspicious logins, phishing, anomalous network traffic) using Splunk Enterprise, collaborating with IT teams on remediation.
- **CyberSecurity Intern (Full-time)** – *loanDepot* *Summer 2022*
  - Designed and deployed phishing simulation campaigns to educate employees on social engineering threats. Gained hands-on exposure to incident response procedures, compliance auditing, and security architecture.

Projects

- **Flora – Senior Capstone (Bloomfilter.ai):** Year-long senior group project working with Bloomfilter improving HCI design for AI agent integration in project management tools, focused on increasing user adoption in enterprise software.
- **Zedi – Personal Finance SaaS:** Built and deployed full-stack finance platform using Rust/Axum REST API, Next.js/TypeScript frontend, PostgreSQL with row-level multi-tenancy, Claude Vision receipt OCR, and Stripe billing, shipped on Railway. (zedi.sneeks.ai)
- **HomeLab Infrastructure:** Self-hosted Proxmox cluster running 12+ services across pfSense networking, Docker, and ZFS mirrored storage, including ClickHouse, Grafana, n8n, Pi-hole, Tailscale, WireGuard, Bitwarden, Plex, and Cloudflare Tunnels, with fleet-wide SSH automation.

Leadership

- **Cal Poly Club Baseball** – Co-President (June 2025 – June 2026), Player (2023 – June 2026)
- **Kappa Sigma Fraternity** – President (Dec 2022 – Mar 2023), Risk Manager (Sept 2021 – June 2025)

Skills

Languages: Python, SQL, Rust, TypeScript, JavaScript, Swift/SwiftUI, Bash, YAML, HTML/CSS
Technologies: Node.js, React, Next.js, Streamlit, PyTorch, LangChain, PostgreSQL, MongoDB (NoSQL), Firebase, Supabase, Docker, Proxmox, Terraform, Git, CI/CD, Linux, Splunk, Splunk SOAR, Snowflake, dbt, Tableau, Sigma Computing, ClickHouse, Grafana, GCP, Cloudflare, Railway, Stripe, MCP